6. Consider a causal continuous-time LTI system whose input $x(t)$ and output $y(t)$ are related by the following differential equation:

$$\frac{d}{dt}y(t) + 4y(t) = x(t).$$

Find the Fourier series representation of the output $y(t)$ for the input $x(t) = \cos(2\pi t)$.

$$x(t) = \frac{e^{j2\pi t} + e^{-j2\pi t}}{2} \qquad \omega_0 = 2\pi \Rightarrow a_1 = a_{-1} = 1/2$$

Suppose:

$$x(t) = e^{j\omega t} \Rightarrow e^{j\omega t} H(j\omega) \cdot j\omega + 4 e^{j\omega t} H(j\omega) = e^{j\omega t}$$

$$\Rightarrow H(j\omega) = \frac{1}{4 + j\omega}$$

$$x(t) = \sum_k a_k e^{jk\omega_0 t}$$

$$y(t) = \sum_k a_k H(jk\omega_0) e^{jk\omega_0 t}$$

$$k = \pm 1 \Rightarrow y(t) = a_1 H(2\pi j) e^{j\omega_0 t} + a_{-1} H(-2\pi j) e^{-j\omega_0 t}$$

$$y(t) = \frac{1}{2}\frac{1}{4 + 2\pi j} e^{j\cdot 2\pi t} + \frac{1}{2}\frac{1}{4 - 2\pi j} e^{-j\cdot 2\pi t}$$

$$y(t) = \frac{(2 - \pi j) e^{j\cdot 2\pi t}}{16 + 4\pi^2} + \frac{(2 + \pi j) e^{-j\cdot 2\pi t}}{16 + 4\pi^2}$$

$$= \frac{2}{16+4\pi^2}\left(e^{j\cdot 2\pi t} + e^{-j\cdot 2\pi t}\right) \quad \frac{-\pi j}{16+4\pi^2}\left(e^{j\cdot 2\pi t} - e^{-j\cdot 2\pi t}\right)$$

$$= \frac{\cos(2\pi t)}{\pi^2+4} + \frac{\pi \sin(2\pi t)}{8+2\pi^2}$$

$$y(t) = \frac{2\cos(2\pi t)+\pi\sin(2\pi t)}{2\pi^2+8}$$